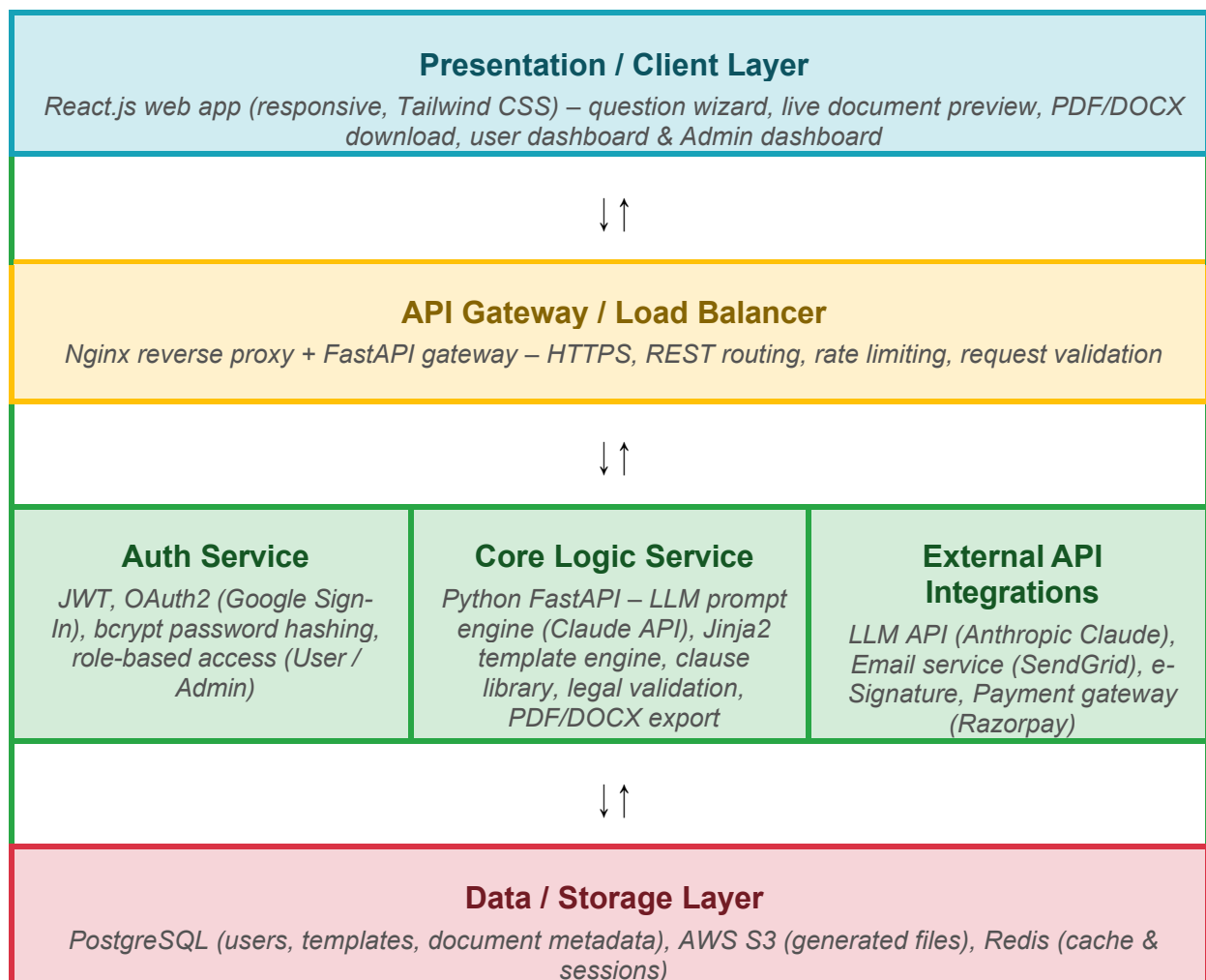


Solution Architecture

Date	1 October 2026
Team ID	SWTID-2026-6275
Team Name	Cipher
Team Members	U. Harikrishnan (Leader), N. Manikandan, B. Sri Harani Devi, K. Kesavan
Project Name	Legal Ease: AI-Powered Legal Document Generator
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	User-facing interface where users choose a document type, answer guided questions, preview the AI-generated draft, edit it and download it. Includes an Admin dashboard to manage templates and users.	React.js, HTML5, CSS3, Tailwind CSS, Axios
API Gateway	Single entry point for all client requests. Handles routing, HTTPS/SSL, rate limiting, input validation and forwards requests to the correct backend service.	Nginx, FastAPI, REST APIs, JSON
Microservices	Auth Service (login, signup, tokens); Core Logic Service (builds prompts, calls the AI model, fills templates, checks mandatory clauses, exports PDF/DOCX); Integration Service (email, e-sign, payments).	Python, FastAPI, Claude API, Jinja2, python-docx, ReportLab, JWT, OAuth2
Database	Stores user accounts, document templates, clause library, generation history and saved drafts securely; generated files are kept in object storage.	PostgreSQL, AWS S3, Redis